FEYNMAN AEROSPACE INTERNSHIP REPORT BATCH 2 (2024-25)



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Table of Acronyms

Sr. No.	Acronym	Term
1	MAV	Micro Aerial Vehicle
2	C_L	Coefficient of Lift
3	C_D	Coefficient of Drag
4	AOA	Angle of Attack
5	AR	Aspect Ratio
6	MAC	Mean Aerodynamic Chord

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ABSTRACT:

The goal of this project was to design a fixed wing micro class RC aerial vehicle in compliance with the rules set by SAE VIT Aerodominator 2020 Competition. This aircraft was not built in reality but the report includes the manufacturing process that could be used to build this aircraft as well as the overall design, analysis and performance. The design fulfils the objective of creating an unmanned aerial vehicle capable of transporting the maximum possible payload, in this case medical supplies, while also being able to achieve good performance. The modelling was done using Fusion 360 and the analysis was done using XFLR5.

I. INTRODUCTION

The primary objective of this project was to design, analyze and possibly build a MAV, understanding the aerodynamics involved and choosing the correct motors, controllers for its operation. To achieve the said objective, extensive research was done in determining the various parameters for building of the plane. Based on this research, it was decided to develop a plane, such that we maximize the aspect ratio, minimize the wing loading and optimize the weight.

The project was completed by a team of five students. Each student was assigned a different task and every student was responsible for thorough research regarding their tasks. Once the initial research was complete, all the members worked together on various parts of the project.

II. AIRFOIL SELECTION

An airfoil shaped body, moving through a fluid, produces aerodynamic forces, which are lift and drag and it is the airfoil geometry which determines the chord-wise lift distribution of a wing. Therefore, one of the important design phases of making an aerodynamically efficient MAV is the selection of the appropriate airfoil.

Airfoil selection of our wing design required performing aerodynamic performance analysis of different airfoils to compare according to the requirements which demanded a "high lift, low Reynold's number" airfoil. The three airfoils considered were S1223, FX63-137 and E423 airfoils and their analysis was done using XFLR5 software. The properties of these airfoils were studied which is tabulated below and points were given accordingly (5 being the best and 1 being the worst) to determine the best airfoil for this aircraft.

Airfoil	C_{Lmax}	**α**	C_L/C_D **at α**
S1223	2.28	13	73 at 4 degrees
FX63-137	1.68	15	84 at 7.5 degrees
E423	1.99	12	63 at 4.5 degrees

Table 1: Airfoil Properties

Airfoil	C_L	a_{stall}	C_L/C_D	**Total**
S1223	5	3	3	11
FX63-137	1	5	5	11
E423	3	2	2	7

Table 2: Point Based System for Airfoil Selection

The C_L behavior of these three airfoils is shown in the graph below:

- **Purple is FX63-137**
- **Yellow is S1223**
- **Red is E423**

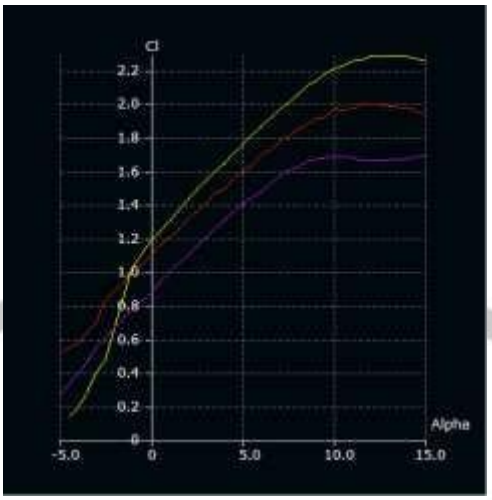


Figure 1: C_L Characteristics of S1223, FX63-137 and E423 Airfoil

There was a tie between the S1223 and FX63 -137 airfoils and although the S1223 airfoil has greater lift characteristics, the FX63 -137 airfoil was chosen to be the suitable airfoil considering the greater tendency of the S1223 airfoil to stall. The FX63 -137 provided a maximum coefficient of lift of 1.7 at an angle of attack of 10 degrees. In order to generate more lift during take -off, an initial angle of attack of 5 degrees was provided to the wing.

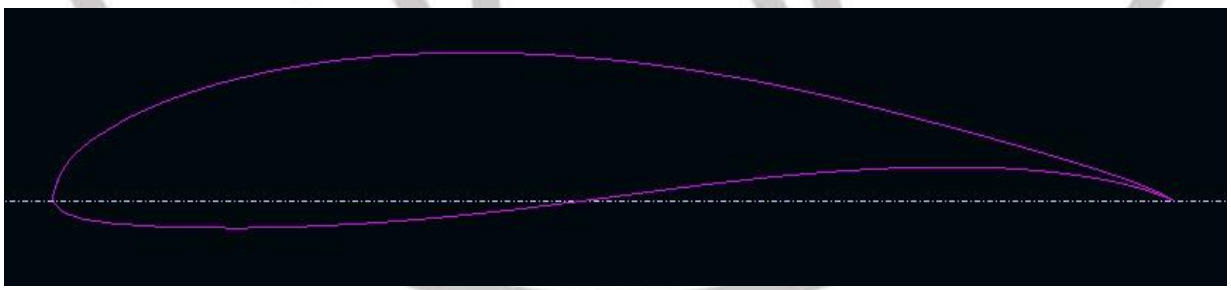


Figure 2: FX63-137 Airfoil

III. WING CALCULATION AND PLANFORM SELECTION

To calculate the total lifting area, we used the formula:

$$\text{Wing Loading} = \frac{\text{Maximum Take} - \text{of Weight}}{\text{Lifting Area}}$$

The most optimum value of wing loading was selected for calculating the lifting area. After considering various values of wing loading ranging from 8 to 12, a wing loading of 10 kg/m^2 was selected and the lifting area to lift a weight of 1.3kg (assumed aircraft weight) was found to be $1300 cm^2$.

Various aspect ratios were considered and our desired range was between 5 – 6.5 according to the requirements of this aircraft. Considering the maximum wing span of 85cm according to the competition rules, and using the previously calculated wing area of $1300cm^2$, the aspect ratio, AR, was calculated using the formula:

$$\text{AR} = = 5.5 \frac{\text{span}^2}{\text{wing area}}$$

The MAC was calculated using the formula:

$$\text{MAC} = \frac{\text{span}}{\text{AR}} = 15.45 \text{ cm}$$

Keeping this in mind, a compound planform was chosen to achieve the best possible lift with a span of 85cm, root chord of 21cm and a tip chord of 10cm. We also added a root dihedral angle to the wing since it helps in easy handling of the aircraft, reduces roll there by increasing longitudinal stability. Shoulder wings were chosen to avoid wing flex and to avoid any wing damaged during landing and also to avoid cantilever stresses on fuselage. It was decided that the wing be made of balsa wood because it was convenient and balsa wood provided great strength for its weight and since the balsa wood is laser cut it would provide greater accuracy than the other materials during manufacturing.

IV. FUSELAGE DESIGN

A fair amount of research amount was done for the fuselage design and it was found that for most RC plane a proportional length of fuselage is 80% - 85% of the wingspan. We also came to a conclusion that manufacturing a stream line design would be difficult therefore, so it was decided

that we would be creating a boxy fuselage as shown below.



Figure 3: Fuselage Isometric View

The material that we decided to use for the fuselage was balsa wood. It was chosen over other allowed materials because it is light in weight and provided sufficient strength.

V. WING ANALYSIS

Wing analysis was done using XFLR5 software. A VLM2 analysis was done using a fixed speed polar where the free stream velocity of operation was taken to be 15m/s along with the analysis conditions given in the table below.

Table 3: Wing Analysis Conditions

Parameter	Value
Speed	15m/s
AoA	-5deg to 15deg

Air Density	$1.225\ kg/m^3$
Kinematic Viscosity	$1.5x10^5\ m^2/s$

The analysis showed that the wing generated a maximum coefficient of lift of 1.67. The variation of the coefficient of lift with angle of attack is shown below:

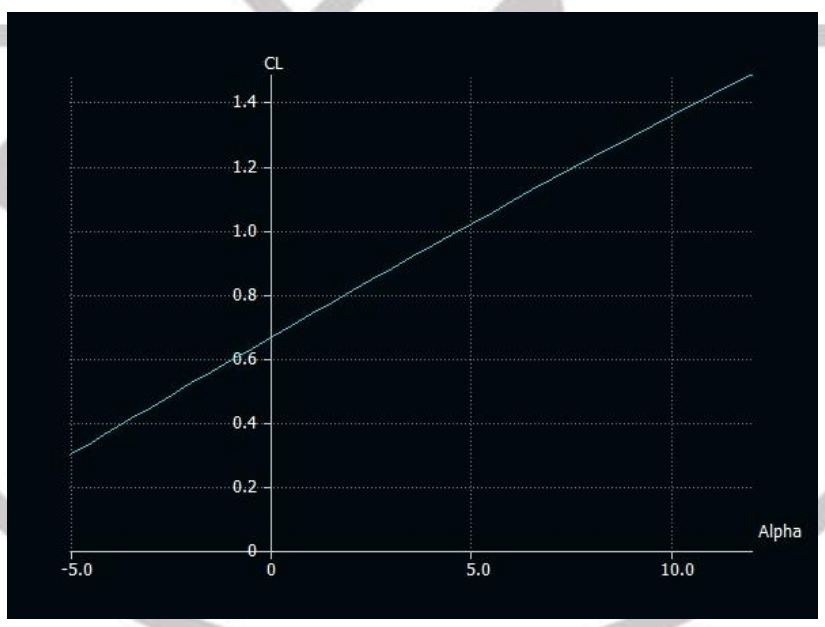


Figure 4: The variation of the coefficient of lift with angle of attack

The variation of C_L/C_D with angle of attack is shown below:

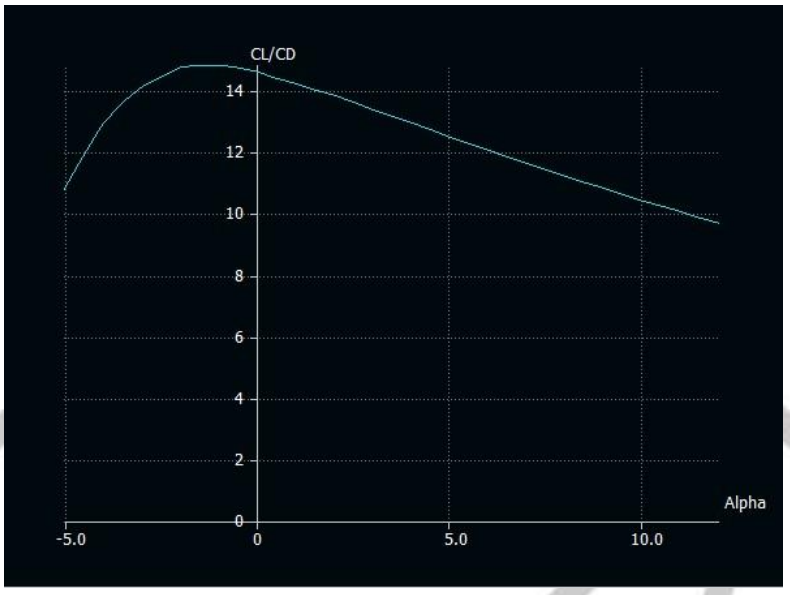


Figure 5: The variation of C_L/C_D with angle of attack

The analysis showed our wing planform produced very good lift within an angle of attack range of -5 to 15 degrees and the maximum lift coefficient generated by the wing was 1.67 at an angle of attack of 15 degrees.

VI. AVIONICS

The avionics system was selected based on the appropriate thrust and design required for the take-off since the plane had to be hand launched. To achieve these goals, it was crucial to select a combination of BLDC motor and propeller which provided a good enough thrust.

The motor used for this aircraft is a Brushless Direct Current Motor. It was c hosen over a regular brushed motor since it provided better overall efficiency and torque per voltage. The bump in efficiency would be especially useful in increasing the range of the plane, and the torque advantage would clearly help in maximizing thrust, and consequently payload capacity.

The motor is a 1400kV A2212 motor having a stator diameter of 22mm and 12 stator arms. The 1400kV rating means the motor should ideally run at 1400rpm for every 1V of voltage applied across it. The motor was found to be 80% efficient meaning it ran at around 1100rpm per volt. The propeller used is an 8085R propeller having two blades, an 8" prop length and a pitch of 45 degrees in the right-hand side. This configuration was chosen over others since it provided a good balance between stable and nimble flight and also provided a good enough thrust for the given payload.

The motor is powered using an 11.1V 3s lithium polymer battery with a 2200mAh capacity. This proved to be a good balance between weight and capacity. Any higher, and gains in the range would be rendered useless by the drop-in performance from the extra weight of the bigger pack.

The servos used are 9gram SG90 servos with plastic gears. These were chosen over the metal geared counterparts to avoid the weight penalty, and also because the forces acting on the surfaces weren't strong enough to exceed the limits of the plastic ones.

The stall torque on each was about 2.5 kg-cm. This is well above the measured forces from the ailerons, the elevators, the rudder, and the payload release mechanism. A single servo mounted right under the wing actuates both ailerons using a 2cm horn that was linked to them via a 2:1 ratio linkage making flight corrections easy and smooth. It also got rid of the twitch that was experienced with a 1:1 linkage.

Two other servos mounted towards the end of the tail are used to control the elevator and the rudder. Both using 2cm horns, with a 180-degree rotation limit. Mounting the servos as rearward as possible helped relocate the center of gravity to underneath the wing shoulder. A fourth servo unhooks the payload doors at the bottom for dropping the payload. All of this is controlled using a 6- channel radio transmitter and controller. Since the payload is delivered mid-flight, there was no need for an extensive landing gear setup.

VII. Payload Drop Mechanism

The payload is delivered via a door at the bottom portion of the aircraft. The payload is stored in a compartment inside the fuselage underneath the ESC. A servo motor is responsible for opening and closing the said doors using a 1:1 ratio linkage. This servo only has a 90-degree rotation limit. Since the doors don't need to open backwards.

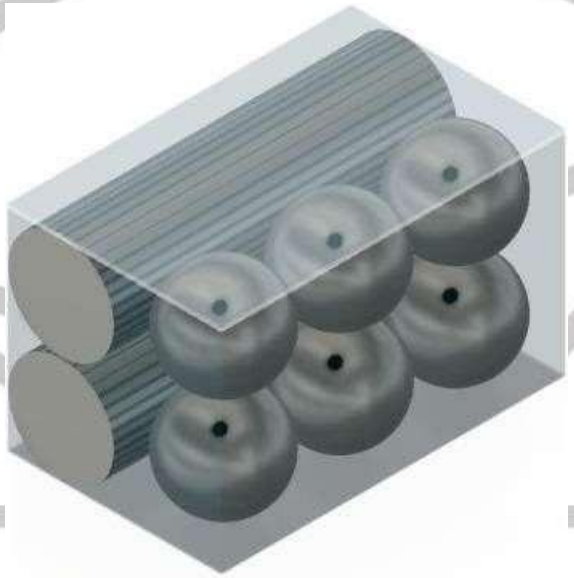


Figure 6: Payload Capsule

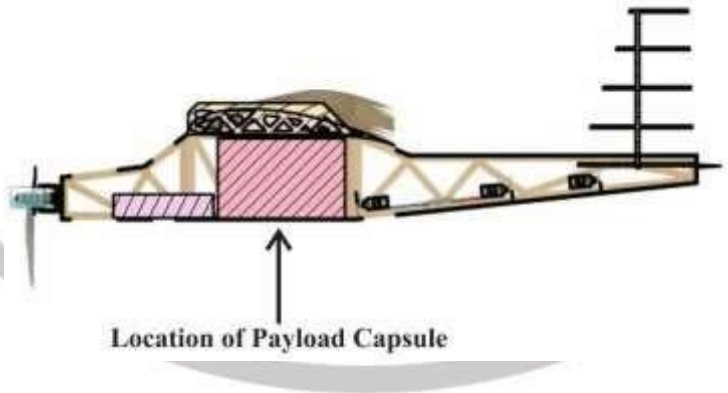


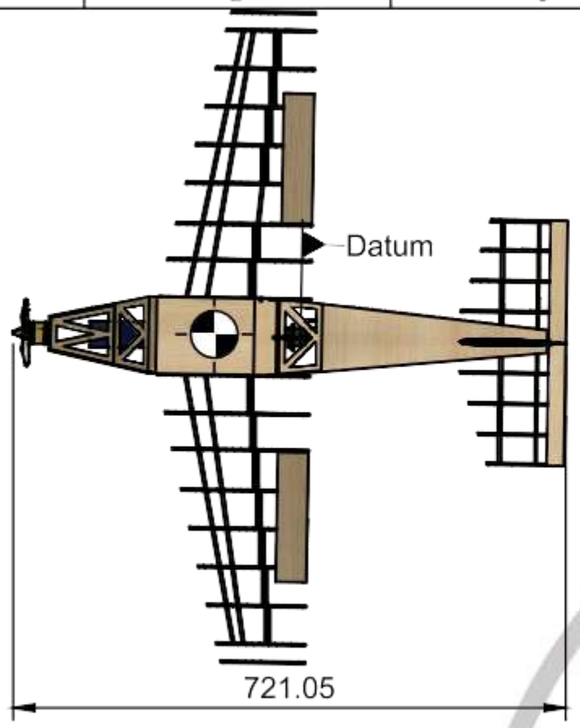
Figure 7: Location of Payload

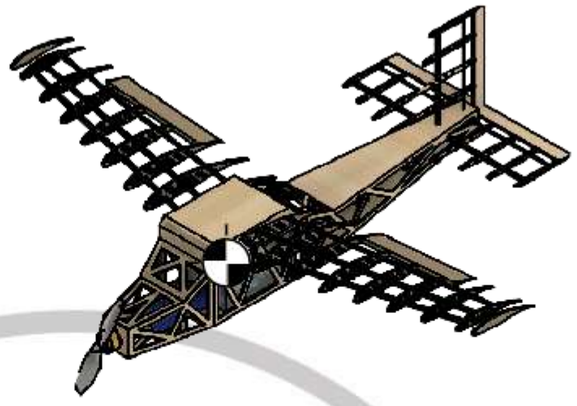
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2	Akshay Prasad & Dr. Nitin Sherje	Design of Regular Class Aerial Vehicle

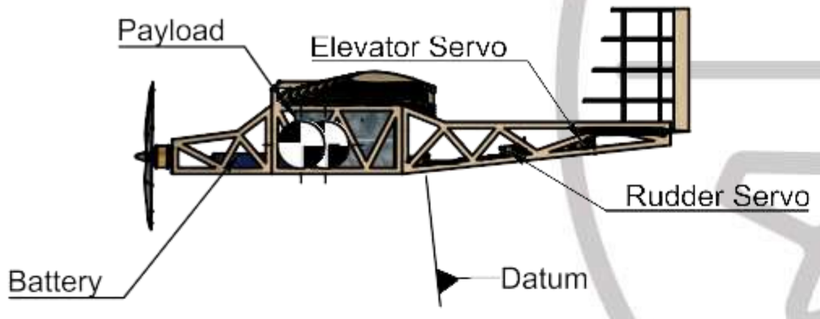
3	Preet sambare	Model Aircraft Design
4	Prashant yadav	R/C Model Aircraft Design

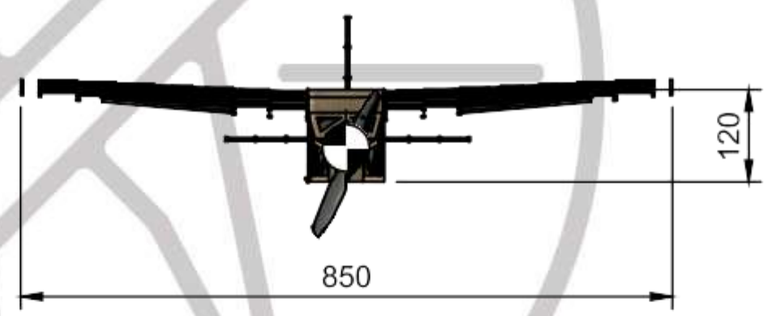






Aircraft MAC	165 mm
Empty Weight CoG Position	-162.94 mm
With Payload Cog Position	-133.14 mm





Wing Span	850mm
Empty Weight	800 gm
Motor	1400KV
Propeller	8045
ESC	30A
Servo	SG90 1200gm.cm
Battery	2200mAh

Tail	289.97 mm
Wing	-61.66 mm
Fuselage	-80.25 mm
Motor	-343.72 mm
ESC	61.671 mm
Battery	-223.89 mm
Payload	-103.54 mm

Team Name	Team Peregrine
Team No.	AD-008
College Name	M.H.S.S.C.E

Micro Class Tech-Data Sheet: Weight Build-Up

Team Name: Team Peregrine Falcons

Sr. No.	Components	Weight (g)
1	Wing	63
2	BLDC Motor	53
3	Propeller	9
4	Fuselage	100
5	Servo	11 X 4
6	ESC	6.5
7	Tail	20
8	Battery	230
9	Mounts, Bolts, Connection Cables, Horns	80
10	Surface	200
Total		805.5